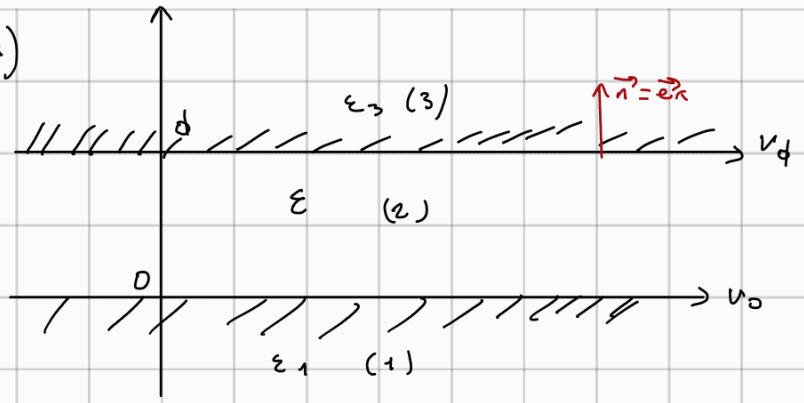
a)



Ara bölgede potansiyel fonksiyonunu, iletkenler üzerinde birim alanda toplanan yükleri bulunuz. $x = d$'de birim alanda toplanan yük Q_d ise $C = Q_d/(V_d - V_0)$'ı bulunuz

Ara bölgede

$$\Delta V = 0$$

$$\frac{\partial^2 V}{\partial x^2} = 0 \qquad V(x) = Ax + B$$

$$x = 0 \Rightarrow V_0 = A \cdot 0 + B$$

$$B = V_0$$

$$x = d \Rightarrow V_d = Ad + V_0$$

$$A = \frac{V_d - V_0}{d}$$

O halde $\boxed{V(x) = \frac{V_d - V_0}{d} x + V_0}$

$$\vec{E} = -\text{grad } V = -\frac{V_d - V_0}{d} \vec{e}_x$$

$x = d$ düzleminde

$$\varepsilon_3 E_n^{(3)} - \varepsilon E_n = \rho_{s,d}$$

($E_n^{(3)} = 0$)

İletken ortamda Elektrik alan 0

$$\rho_{s,d}\, d = \frac{\varepsilon}{d}(V_d - V_0)$$

$$Q_d = \int_S \rho_{s,d}\, ds = \frac{\varepsilon}{d}(V_d - V_0)$$

İletkenler kapattığı bölgenin içinde olan elektrik alanı dışarıya geçirmez.

$$C = \frac{Q_d}{V_d - V_0} = \frac{\varepsilon}{d}$$

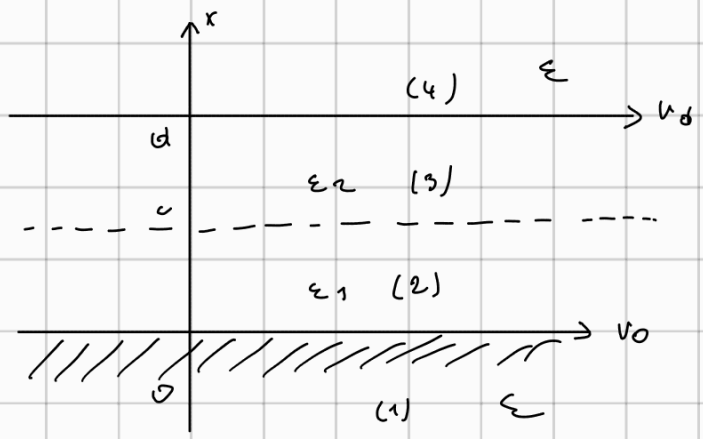
$x = 0$ düzleminde

$$\varepsilon E_n^{(2)} - \varepsilon_1 E_n^{(1)} = \rho_{s,0}$$

($E_n^{(1)} = 0$)

$$\rho_{s,0} = -\frac{\varepsilon}{d}(V_d - V_0) = Q_0 \quad \text{(Birim alanda)}$$

b)



Ara bölgede $\Delta V = 0$

$$V(x) = \begin{cases} Ax + B, & 0 < x < c \\ Dx + F, & c < x < d \end{cases}$$

$x = 0$

$V(0) = V_0$

$B = V_0$

$V(d) = V_d$

$Dd + F = V_d \Rightarrow F = V_d - Dd$

$x = c \quad V^{(2)}(c) = V^{(3)}(c)$

$Ac + V_0 = Dc + F$

$Ac + V_0 = Dc + V_d - Dd$

$Ac + V_0 = D(c-d) + V_d$

$x = c$

$\varepsilon_2 E_n^{(3)} - \varepsilon_1 E_n^{(2)} = 0$ ← yüzey yok

$\varepsilon_1 \frac{\partial V^{(2)}}{\partial x} - \varepsilon_2 \frac{\partial V^{(3)}}{\partial x} = 0$

$\varepsilon_1 A = \varepsilon_2 D \Rightarrow D = A \frac{\varepsilon_1}{\varepsilon_2}$

$Ac + V_0 = A \frac{\varepsilon_1}{\varepsilon_2}(c-d) + V_d$

$$A = \frac{\varepsilon_2 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2 - \varepsilon_1)}$$

$$D = \frac{\varepsilon_1 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2 - \varepsilon_1)}$$

$$F = V_d - D_{ol} = \frac{c\,V_d(\varepsilon_2-\varepsilon_1) + \varepsilon_1 d V_0}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)}$$

$$V(x) = \begin{cases} \dfrac{\varepsilon_2 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)} x + V_0 & , 0 < x < c \\ \dfrac{\varepsilon_1 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)} x + \dfrac{c\,V_d(\varepsilon_2-\varepsilon_1) + \varepsilon_1 d V_0}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)} & , c < x < d \end{cases}$$

$$\vec{E}(x) = \begin{cases} -\dfrac{\varepsilon_2 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)} \vec{e}_x & , 0 < x < c \\ -\dfrac{\varepsilon_1 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)} \vec{e}_x & , c < x < d \end{cases}$$

$x=0$ ve $x=d$ deki yük yoğunlukları

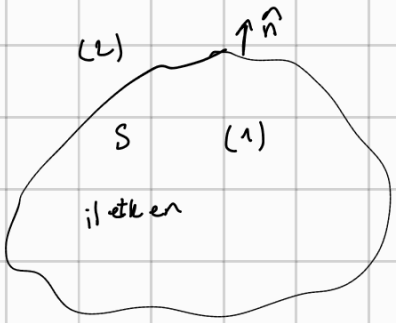
$x=0$ $\quad \rho_{s,0} = \varepsilon_1 E_n^{(2)} - \varepsilon E_n^{(1)}$ ($E_n^{(1)} \to 0$)

$$\rho_{s,0} = -\frac{\varepsilon_1 \varepsilon_2 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)}$$

$x=d$ $\quad \rho_{s,d} = \varepsilon E_n^{(4)} - \varepsilon_2 E_n^{(3)}$

$$\rho_{s,d} = \frac{\varepsilon_1 \varepsilon_2 (V_d - V_0)}{\varepsilon_1 d + c(\varepsilon_2-\varepsilon_1)}$$

SSO PrS



cismin dışında V potansiyeli biliniyor olsun. S üzerinde toplanan yük:

$$Q = -\int_S \varepsilon \frac{\partial V}{\partial n} ds$$

S yüzeyinde

$\varepsilon E_n^{(2)} - \varepsilon_1 E_n^{(1)} = \rho_s$ ($E_n^{(1)} \to 0$)

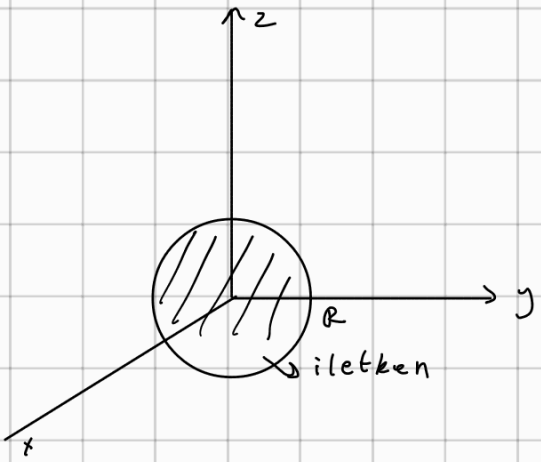
İletkenlerin sadece yüzeyinde yük biriktiği

$$\rho_s = \varepsilon E_n^{(2)}$$

$$\rho_s = -\varepsilon \frac{\partial V}{\partial n}$$

$$Q = \int_S \rho_s \, ds = -\int_s \varepsilon \frac{\partial V}{\partial n} \, ds$$

5.90 Pr6



küre üzerinde potansiyel V_0'dır

küre üzerindeki Q gönünü

ve $C = \frac{Q}{V_0}$' ı bulunuz

$$\Delta V = 0$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{dV}{dr}\right) = 0 \neq V(r) = \frac{A}{r} + B$$

$$r \to \infty, \; V(r) \to 0 \Rightarrow B = 0$$

$$r = R, \; V(R) = V_0 \Rightarrow \frac{A}{R} = V_0 \Rightarrow A = V_0 R$$

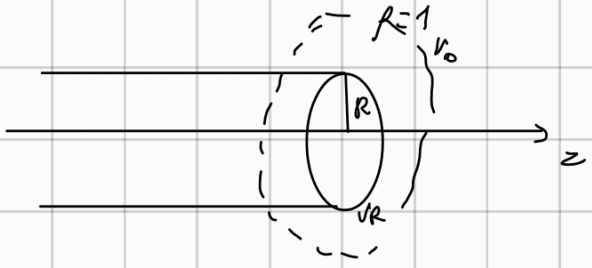
$$V(r) = \frac{V_0 R}{r}$$

$$Q = -\int_S \varepsilon \frac{\partial V}{\partial n} \, ds = +\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \varepsilon \frac{V_0 R}{r^2} \underbrace{r^2 \sin\theta \, d\theta \, d\phi}_{ds}$$

$$Q = 4\pi \varepsilon V_0 R$$

$$C = \frac{Q}{V_0} = 4\pi \varepsilon R$$

5.90 Pr7



Silindirin birim uzunluğundaki Q yükünü ve $C = \frac{Q}{V_0}$'ı bulunuz

$\rho > R$

$\Delta V = 0 \qquad V(\rho) = A \ln\rho + B$

$\rho = 1, \quad V_0 = A \ln 1 + B \Rightarrow B = V_0$

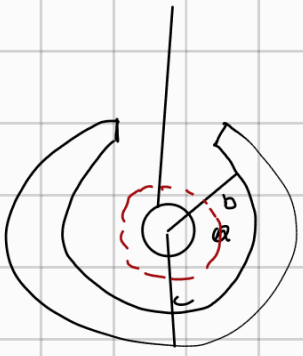
$\rho = R, \quad V_R = A \ln R + V_0 \Rightarrow A = \frac{V_R - V_0}{\ln R}$

$$V(\rho) = \frac{\ln(\rho)}{\ln R}(V_R - V_0) + V_0$$

$$Q = -\int_S \varepsilon \frac{\partial V}{\partial n} ds = -\varepsilon \int_{z=0}^{1} \int_{\phi=0}^{2\pi} \frac{(V_R - V_0)}{\rho \ln R} \rho \, d\phi \, dz$$

$$Q = -2\pi\varepsilon \frac{(V_R - V_0)}{\ln R} \Rightarrow C = \frac{Q}{V_0} = -\frac{2\pi\varepsilon(V_R - V_0)}{V_0 \ln R}$$

S.50 P, 8



Çeper bağlantısı nötr

A yarıçaplı küreçikve çeper iletkenler

a) Oyuk içindeki Elektrik alanı bulunuz

$$\int_S \vec{D} \cdot d\vec{S} = Q$$

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \varepsilon E_r r^2 \sin\theta \, d\phi \, d\theta = Q$$

$$\vec{E_r} = \frac{Q}{4\pi\varepsilon r^2} \Rightarrow \vec{E} = \frac{Q}{4\pi\varepsilon r^2} \vec{e}_r$$

b) kabuğun iç ve dış yüzeylerinde biriken yükler nedir?

$r = b$

$$\rho_{s,b} = \varepsilon_1 E_n^{(2)} - \varepsilon E_n^{(1)} \quad (E_n^{(2)} = 0)$$

$$\rho_{s,b} = \frac{-\varepsilon Q}{4\pi\varepsilon b^2} = -\frac{Q}{4\pi b^2}$$

$$\rho_{s,c} = -\frac{\varepsilon Q}{4\pi\varepsilon b^2} = -\frac{Q}{4\pi b^2}$$

$$Q_b = \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} -\frac{Q}{4\pi b^2} b^2 \sin\theta \, d\phi \, d\theta = -Q$$

$$Q_c = -Q_b = Q$$

c) $r>c$ iken elektrik alan nedir??

$$\int_S \vec{D}\, d\vec{s} = Q + Q_b + Q_c = Q$$

$$\vec{E} = \frac{Q}{4\pi\varepsilon r^2}\vec{e}_r$$

d) Kepere dokundurup çıkarılırsa

Küreçik nötrleşir